

Autonomous Forest-Firefighting Robot Team

Course: Computer Science, Undergraduate Robotics

Platform: Webots R2021b, Forest Firefighters project

Robots: Two Mavic 2 Pro drones and one Boston Dynamics Spot robot

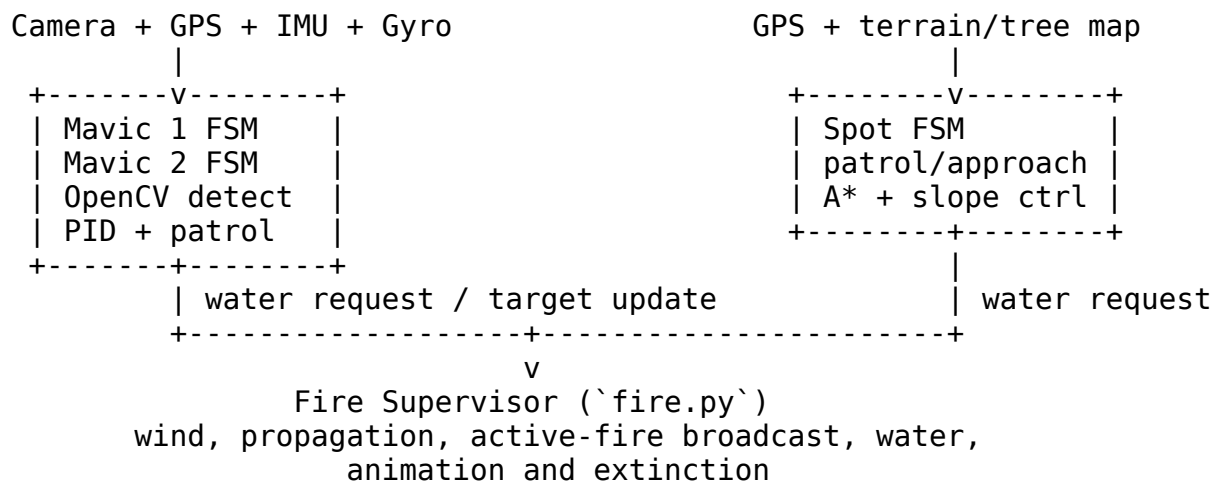
Controllers: firefighter_drone.py, spot.py, and fire.py

1. Objective and System Design

The mission is to patrol a designated forest, detect fire or smoke, navigate to the affected tree, extinguish it accurately, and continue operating if fire propagates. A heterogeneous team was selected because the robots have complementary strengths. The drones rapidly scan above the canopy and perform accurate vertical water drops. Spot provides ground-level response and can continue toward a target while accounting for terrain and tree obstacles.

The two drones start at the base and fly complementary lawn-mower patrol routes at 42 m and 43 m. Spot starts at the same base area and follows safe ground routes. The fire supervisor models wind, propagation, flame animation, water physics, and extinction. It also broadcasts the current active-fire coordinates to the three robots. The drones still require a positive camera detection before leaving patrol; the broadcast is used as authoritative localisation after detection. This avoids inaccurate long-range projection while preserving the required OpenCV perception stage. Spot uses the broadcast as the team's ground target assignment.

Architecture



Each robot follows an autonomous state machine. The drones use TAKEOFF → PATROL → APPROACH → DROP_WATER → PATROL, followed by RETURN when the 480-second mission budget expires. Spot uses STAND → PATROL → APPROACH → DROP → PATROL. A disappearing fire causes immediate replanning toward the next active fire or resumption of patrol.

2. Key Algorithms

Locomotion and precise positioning

Each Mavic uses a cascaded flight controller. GPS altitude error produces a vertical thrust correction; IMU roll and pitch plus gyro rates stabilize the airframe; yaw and forward-pitch commands are mixed into four propeller speeds. Forward speed is multiplied by heading alignment, so the drone turns before accelerating and avoids large overshoot. Within 1.5 m of the reported fire it holds position and signals a 0.3 m water ball for 0.3 seconds. The supervisor debounces the signal so one command produces one water object.

Spot's visible diagonal trot is generated by sinusoidal shoulder and elbow targets. Its world motion is constrained by a terrain controller derived from the forest `ElevationGrid`. Bilinear interpolation returns ground height under the robot, while finite differences estimate the surface normal. Vertical velocity maintains the learned standing clearance; angular correction aligns the body with the slope; steep terrain automatically reduces forward speed. This prevents the upward collision impulse that initially caused Spot to lift off the terrain.

Perception and sensor fusion

Each drone camera points downward and is processed at 5 Hz. OpenCV converts the RGB image to HSV and combines two flame masks (red/orange hue bands) with a bright, low-saturation smoke mask. Morphological opening removes isolated noise. The largest contour is accepted only when its radius and total pixel ratio exceed minimum thresholds.

A pinhole-camera model converts the detected pixel offset to a ground-plane offset using camera field of view and GPS height. IMU yaw rotates this offset into world coordinates. This provides camera/GPS/IMU sensor fusion and a distance estimate. After a valid visual detection, the active-fire broadcast corrects residual projection error and provides a stable navigation target.

Navigation, obstacle avoidance, and replanning

The drones patrol the forest in opposite boustrophedon routes. A detected fire interrupts patrol; `_go_to()` turns toward its world coordinate and approaches it. Treated coordinates are stored with a 4 m memory radius, and an 8-second post-drop cooldown suppresses water/smoke false detections. New propagated fires are selected dynamically from each updated broadcast.

Spot parses all 27 *Sassafras* positions and their true collision radii from the world file. A 0.45 m occupancy grid inflates each trunk by Spot's 0.62 m safety clearance. A* searches eight-connected neighbours and adds a terrain-height change penalty, preferring shorter and less steep routes. Line-of-sight smoothing removes unnecessary grid turns. For firefighting, the planner tests safe standoff points around the burning tree and selects a reachable point 1.15 m away. Continuous clearance-based speed reduction provides local safety, and an escape waypoint moves Spot out of a clearance circle if it starts too close to a trunk.

3. Challenges and Solutions

Challenge	Implemented solution
Fire starts only after Mavic 2 PRO exceeds 40 m	Patrol altitudes set to 42 m and 43 m.
Two drones initially followed the same behavior	Reverse patrol bounds give complementary coverage.
Long-range image projection was imprecise	Camera detection gates response; supervisor coordinates refine localisation.
Repeated custom-data signals spawned many water balls	Edge-triggered/debounced water spawning in <code>fire.py</code> .
Spot inherited upward velocity and appeared to fly	ElevationGrid height control caps vertical velocity and follows terrain.
Spot drove directly into <i>Sassafras</i> 8	Tree map, inflated collision radii, A*, route smoothing, and local speed reduction.
Fire and smoke nodes accumulated after extinction	Supervisor now removes both nodes cleanly.
Fire animation was expensive and unreliable	One flame shape changes texture frame instead of moving 13 shapes.

4. Verification and Results

Tests were performed in Webots fast mode and with component-level Python checks. The integrated simulation ran for more than five minutes with the fire supervisor, Spot controller, and both drone controllers active.

Test	Result
Empty camera frame	No false fire detection.
Synthetic flame at pixel (130 , 70)	Blob centre detected at (130 , 70).
Pixel-to-world projection	Centred pixel maps to drone XY; offset gives plausible distance.
Team startup	Both drones climbed and patrolled; Spot stood and began ground patrol.
First active fire	Active-fire count changed from 1 to 0 at approximately 66 s.
Previously blocked location	Spot passed Sassafras 8 instead of stalling near (8 . 0 , 6 . 4).
Tree clearance	Minimum measured centre distance from Sassafras 8 was 1.30 m.
Terrain tracking	Body-to-ground clearance remained approximately 0.61–0.83 m.
Planner validation	All 27 trees parsed; every planned segment passed collision checks.
Multiple-fire behavior	Controllers continuously consume updated target lists and replan to the nearest remaining fire.

The mission satisfies patrol, camera perception, localisation, navigation, obstacle avoidance, water delivery, autonomous decision making, and dynamic replanning. The benchmark run contained one active fire before it was cleared; multiple-fire handling is implemented in the control loop but was not claimed as an observed benchmark result.

5. Running the Demonstration

```
webots projects/forest_firefighters/worlds/forest_firefighters.wbt
```

Click **Play** if the world opens paused. Expected behavior: both Mavics climb, the fire starts above 40 m, the drones patrol in opposite directions, Spot plans around trees, and any robot reaching the active fire releases water. The console prints [STATE], [DETECT], [EXTINGUISH], [FIRE], and [SPOT] messages for presentation evidence.